An introduction to Logistic Regression

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Examples of two class classification problems

- Predicting stock price movement (up/down)
- Predict whether a patient has diabetes or not
- Predict whether a customer will buy or not
- Predict whether a loan applicant will default

Logistic regression : Overview

- A statistical technique developed by David Cox in 1958.
- Logistic regression is **usually** taken to apply to a **binary dependent variable**, though it can be extended to variables with multiple categories.
- Inputs are one or more predictor variables
- The model itself **outputs a conditional probabilit**y of output in terms of input, however it is usually straightforward to convert this to a classification output.

Background concepts and notation

- Consider a two class classification problem. Let the class labels be coded as 1 and 0 and let 1 denote the target class.

- Let p denote the probability of occurrence of target class, given predictor variables.
  i.e. p = conditional probability P(Y=1 | x) where x could represent one or more predictors.

- The **odds of success** (sometimes just called odds) is defined as :

  odds = p/(1-p)

- Sometimes it is convenient to use log-odds which is defined as the natural log (log to base e) of the odds. This is often referred to as the logit function :

  logit(p) = ln (p/(1-p))

Illustrative examples where logistic regression may be suitable

Example 1 : Predict approval based on credit score

creditScore	approved
655	0
692	0
681	0
663	1
688	1
693	1
699	0
699	1
683	1
698	0
655	1
703	0
704	1
745	1
702	1

$n = 1000$

creditScore is the applicant's credit score

approved is coded "1" for approved and "0" for not approved; it is a binary, mutually exclusive variable.

* Only 15 of 1000 observations shown

Source : Brandon Foltz, Logistic Regression youtube

Aim : Develop a model that takes a given credit score as input and computes the probability (and hence odds) of being approved.

Typical questions one would like to answer using the model

Discover from the data approximately what credit score is associated with a probability of 50% of being approved (odds are even).

Predict from the model how improving credit score from say 720 to say 750 affects the probability of being approved.

Source : Brandon Foltz,Logistic Regression youtube

Example 2 : Auto club mailer flier

An auto club mails a flier to its members offering to send more information regarding a supplemental health insurance plan if the member returns a brief enclosed form.

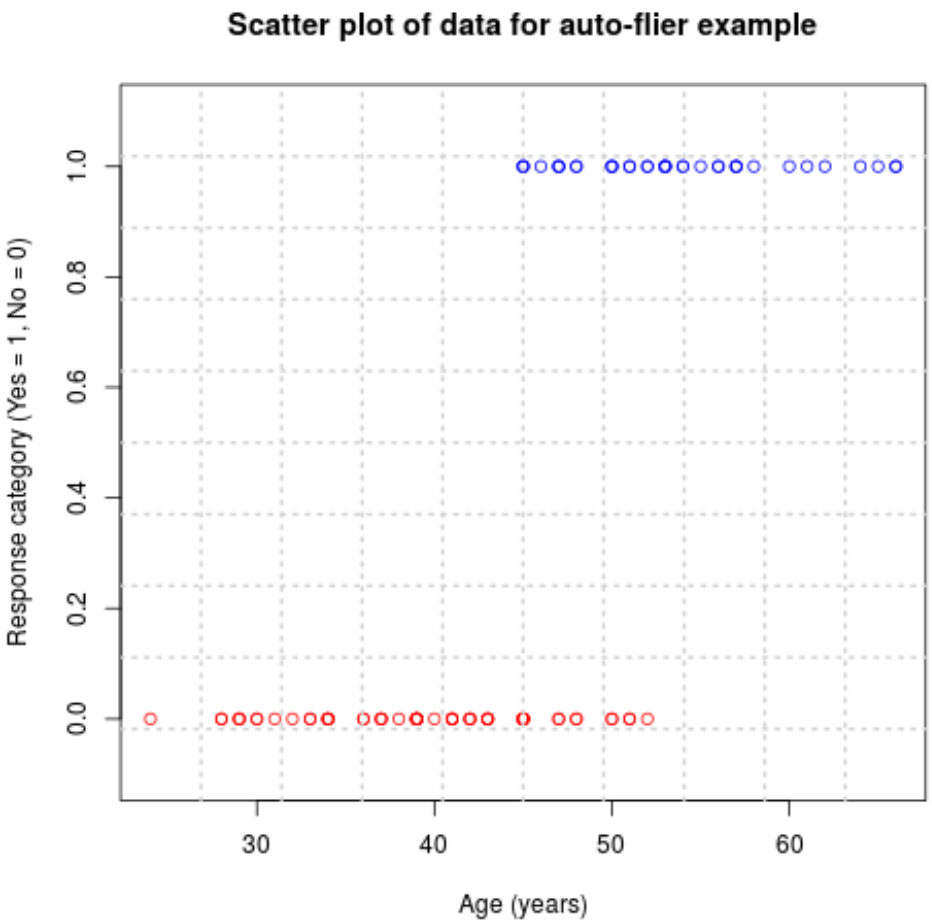
- Can a model be built to predict if a member will return the form or not?
- Additionally, can the model compute the probability (and hence odds) of a particular member returning the form?

Building a logistic regression model

Example 2 : Auto club mailer flier

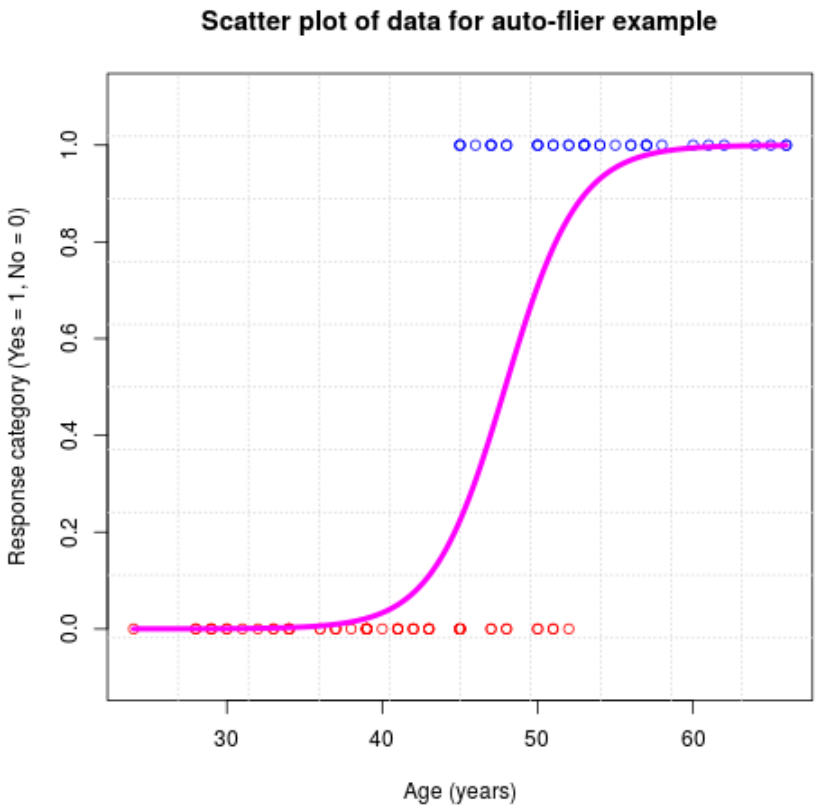
Table of data, only few records shown

Age	Response
...	...
51	1
64	1
54	1
42	0
45	0
33	0
...	...



Response categories coded as
Yes : 1 No : 0

A logistic regression model

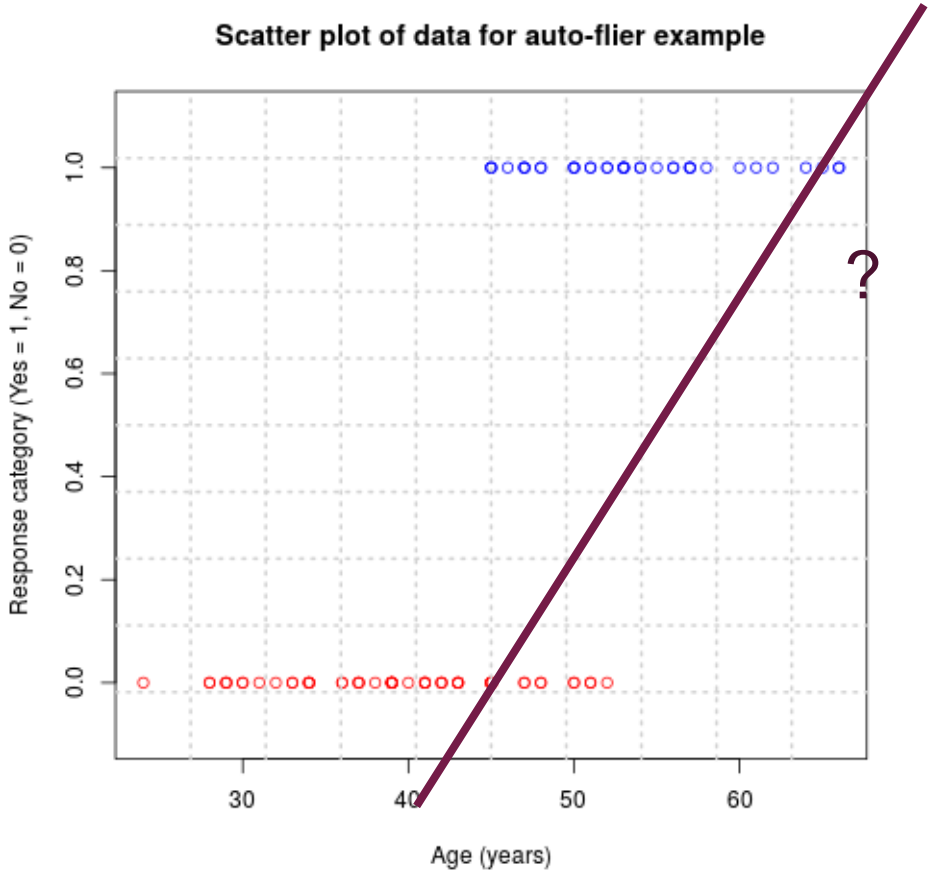


Model inputs :
Value of one or more predictors

Model output :
Prob(Class = Target | predictor values) = $P(Y=1 \mid x_1, x_2, \ldots, x_k)$

(In our example, Age is the only predictor).

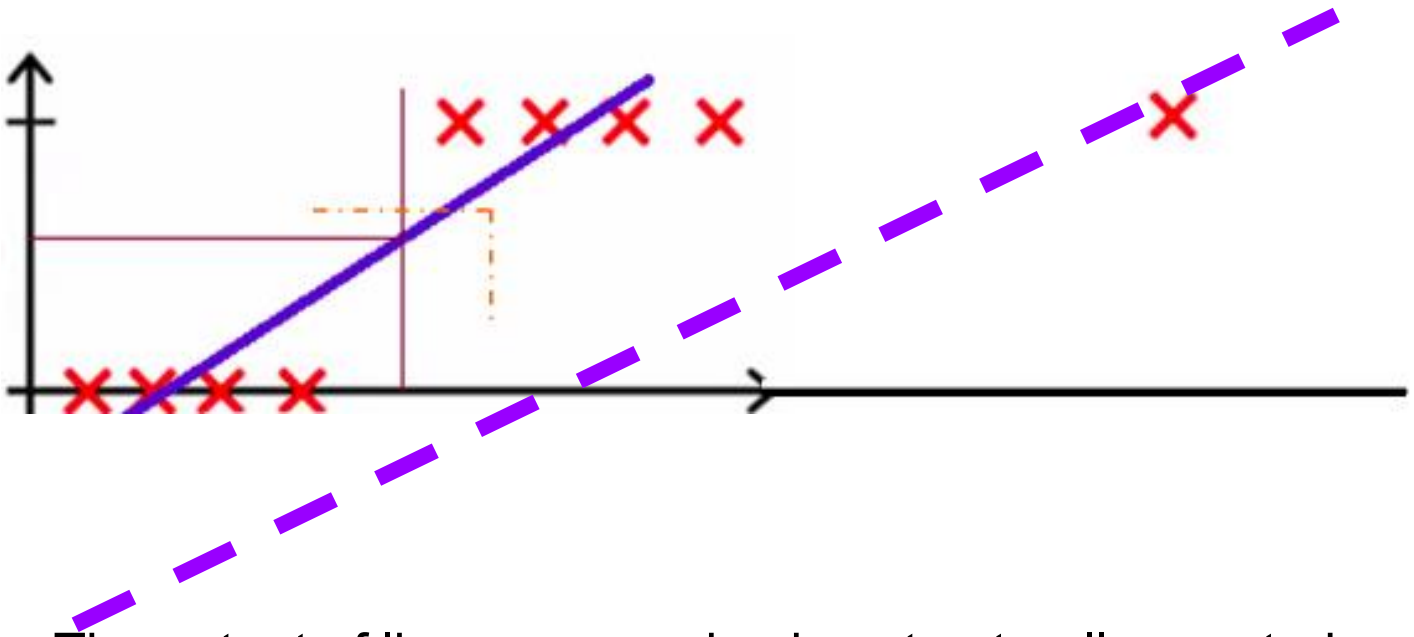
Classification Tasks: Can regression be used?



Consider a two class problem, where the class labels are coded as 0 and 1

How about using linear regression for classification with target values as the class labels 0 and 1?

Linear regression for classification has major drawbacks



- The output of linear regression is not naturally constrained to lie within a range.
- Linear regression slopes can vary significantly depending on the data and hence thresholding becomes difficult.

Other reasons why Linear regression is not suitable

Basic assumptions of linear regression are clearly violated.

- The target variable clearly does not follow a normal distribution (binomial in our case).
- Error terms do not follow normal distribution.
- Error variances are heteroscedastic.

Hence, Linear Regression via Least Squares is inappropriate.

An overview of Generalized Linear Models

$x_1, x_2, \ldots, x_k$

Predictor variables

$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$

Linear combination or a regression with additional terms

Link function

$P(Y=1 \mid x_1, x_2, \ldots, x_k) = p$ (say)

An output which is a probability or an expected value related to the response variable (y)

Logistic regression as a GLM

Predictor variables

Linear combination of predictors

$x_1, x_2, \ldots, x_k$ → $z = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$ → **Link function**

logit function

$$z = \sigma^{-1}(p) = ln\left(\frac{p}{1-p}\right)$$

$P(Y=1 \mid x_1, x_2, \ldots, x_k) = p$ (say)

where σ represents the sigmoid function

An output which is a probability or an expected value related to the response variable (y)

$$\sigma(z) = \frac{1}{1+e^{-z}} = p$$

Logistic regression and decision boundary

Logistic regression enforces a **linear decision boundary**, specifically,
it computes a decision boundary assuming that
log(Odds ratio) is a linear combination of predictor variables.

i.e. for a binary classification problem if 1 denotes the target class and Y denotes the class label it is assumed that

$$\log(P(Y=1 \mid x) / P(Y=0 \mid x)) = \boldsymbol{\beta}_0 + \boldsymbol{\beta}_1 x_1 + \ldots + \boldsymbol{\beta}_k x_k$$

Logistic regression : Computational aspects

In logistic regression, the parameters are computed by maximizing a log loss function. Commonly, the log of the conditional likelihood $P(\boldsymbol{\beta} \mid x)$ (where $\boldsymbol{\beta}$ is the set of parameters) is maximized, using the training data as observations.

Computation of parameters via maximization of log loss can also be shown equivalent to performing a **Maximum Likelihood Estimation (MLE)**

The loss function used for optimization of logistic regression model parameters is typically convex. The optimization is usually performed using gradient descent (gradient ascent) as appropriate or Newton-Raphson method.

We skip the computational details here. The reader may refer resources listed in references for computational details.

A small detour :
Maximum Likelihood Estimation

Maximum likelihood estimation (**MLE**) is a method of estimating the parameters of an assumed probability distribution, given some observed data.

- Typically this is achieved by maximizing a likelihood function such that the observed data is most probable, under the assumed statistical model,.
- The point in the parameter space that maximizes the likelihood function is called the maximum likelihood estimate.

A simple illustrative example of Maximum Likelihood Estimation

Consider tossing a coin multiple times. Let p_H be the probability of the result of a random coin toss being a Head.
Suppose the result of a series of 10 coin tosses is H,T,H,T,T,H,H,T,H,H.

Exercise : Estimate the parameters of the underlying assumed probability distribution, based on Maximum Likelihood Estimation on the above observed data.

References

A blog with code snippets
https://towardsdatascience.com/logistic-regression-detailed-overview-46c4da4303bc

Lecture notes from Data Mining course taught by Dr. Jia Li, Penn State University
https://www.scribd.com/document/332869746/logit

Lecture notes
https://engineering.purdue.edu/ChanGroup/ECE595/files/Lecture14_logistic.pdf

Course notes of CS229 taught by Andrew Ng
https://see.stanford.edu/materials/aimlcs229/cs229-notes1.pdf